

Indicative content

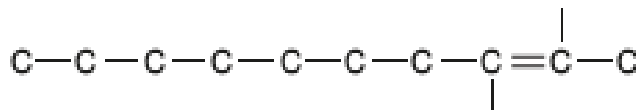
65.2 % of carbon and M_r 184
' M_r ' of the carbon content is $65.2 \times 184 / 100 = 120$
therefore must be 10 carbon atoms

26.1 % of oxygen and M_r 184
' M_r ' of the oxygen content is $26.1 \times 184 / 100 = 48$
therefore must be 3 oxygen atoms

remaining mass 16 therefore must be 16 hydrogen atoms

⇒ molecular formula is $C_{10}H_{16}O_3$

straight chain, double bond between carbons 2 and 3 and an *E*-isomer



2,4-DNP derivative, but no silver mirror therefore must be a ketone

gives the triiodomethane test therefore must contain $\text{CH}_3\text{C}=\text{O}$ group;
these must be carbon atoms 9 and 10

effervescence therefore must be a carboxylic acid; not branched, therefore carbon 1 must be the carboxylic acid group

formula must be

